

The background of the slide features a blurred image of a person in a business suit holding a tablet. Overlaid on this image are several digital and network-themed graphics: a glowing blue grid of data points in the upper right, a network diagram with nodes and lines in the lower right, and a stylized blue circular icon on the tablet screen. The overall color palette is dark blue and teal.

Roy Daher

Docker, Compose and Swarm

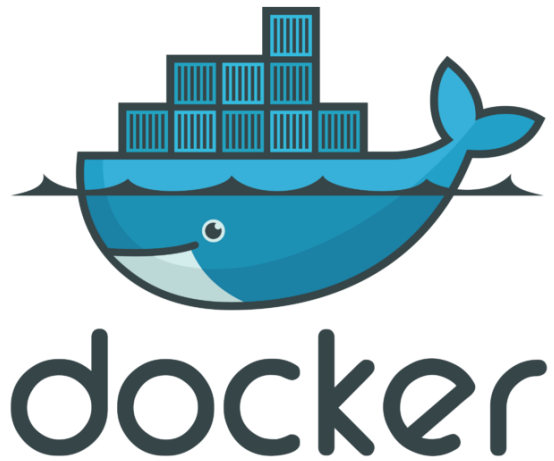
OVERVIEW

- What is Docker
- What is a Container
- Docker Image
- Dockerfile
- Docker-compose
- Docker Swarm
- Conclusion



What is Docker

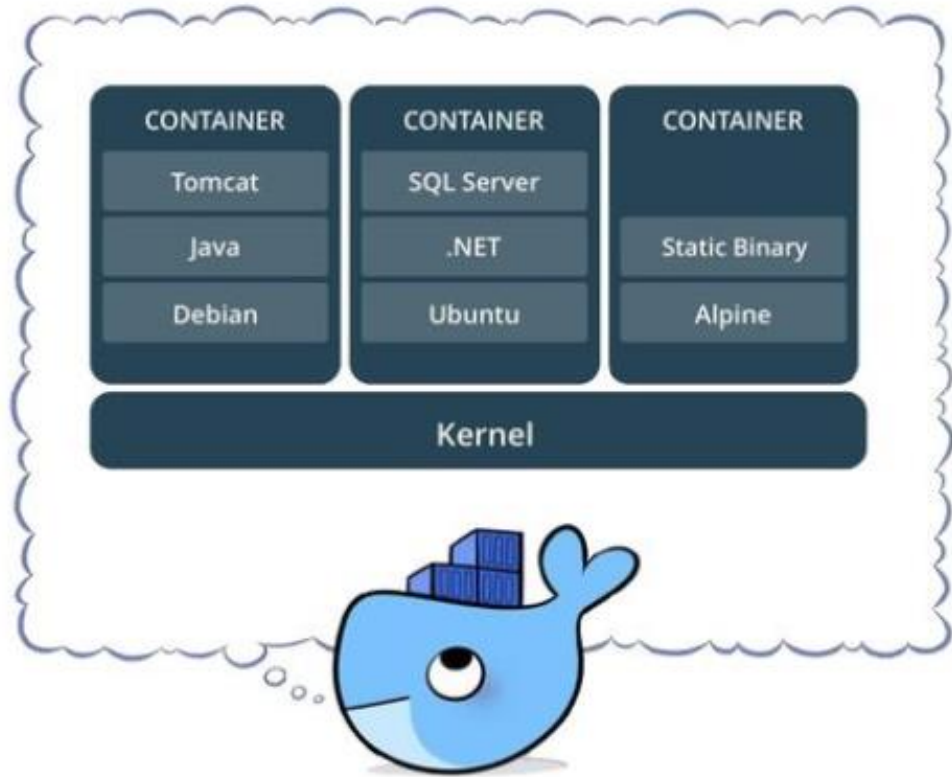




- Lightweight, open, secure platform
- Simply building, shipping, running apps
- Runs natively on Linux or Windows Server
- Runs on Windows or Mac Development machines (with a virtual machine)
- Relies on "images" and "containers"

What is a Container

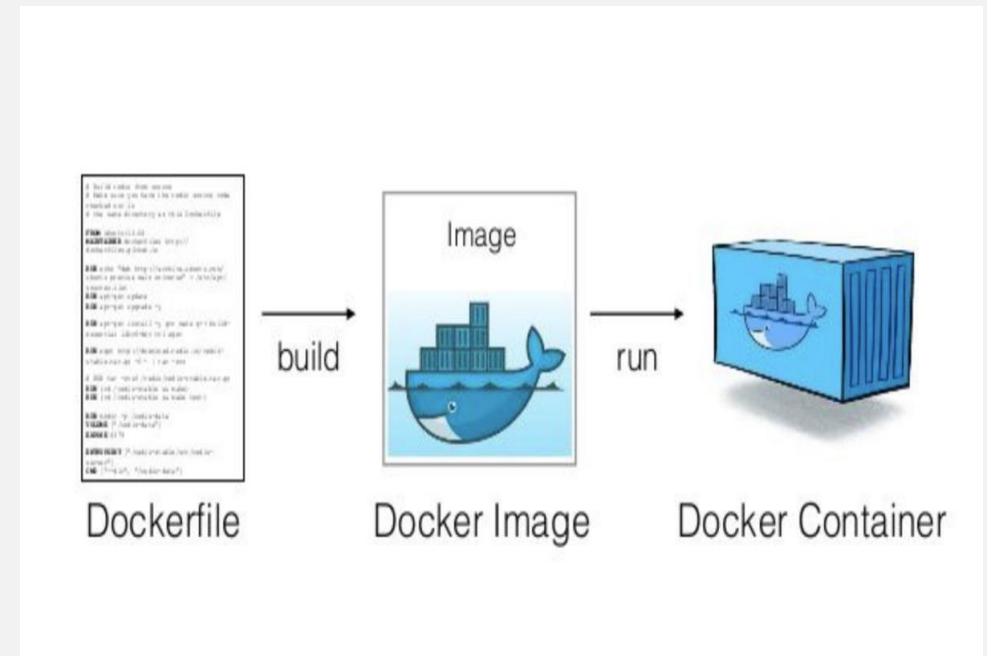




- Standardized packaging for software and dependencies
- Isolate apps from each other
- Share the same OS kernel
- Works for all major Linux distributions
- Containers native to Windows 2016

Docker Image

- Foundation of Docker containers as templates
- Include the whole file system bundle
- Official images are retrieved from open image sources such as Docker Hub
- Whenever an image is run, the resulting filesystems and behaviors are same



Dockerfile

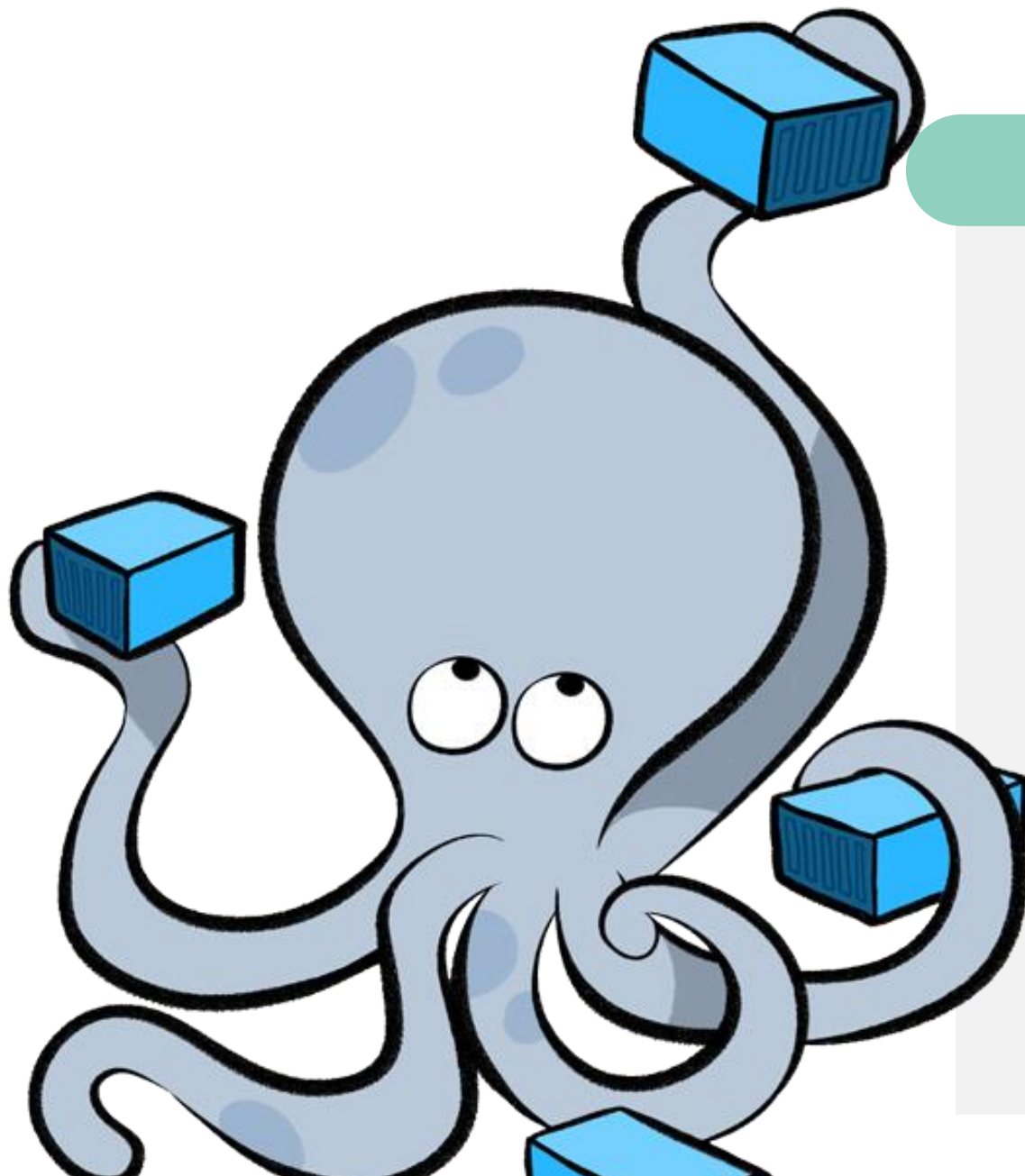
- Command line instructions a user could use to put together an image
- Automates the process of creating images so that they can be shared and replicated across many contexts.
- Dockerfiles specify how an image is stacked and put together, as well as how everything in it connects to one another.
- Distributing the application and its construction instructions as code is made possible using Dockerfiles.

```
### STAGE 1:BUILD ###
# Defining a node image to be used as giving it an alias of "build"
# Which version of Node image to use depends on project dependencies
# This is needed to build and compile our code
# while generating the docker image
FROM node:12.14-alpine AS build
# Create a Virtual directory inside the docker image
WORKDIR /dist/src/app
# Copy files to virtual directory
# COPY package.json package-lock.json ./
# Run command in Virtual directory
RUN npm cache clean --force
# Copy files from local machine to virtual directory in docker image
COPY . .
RUN npm install
RUN npm run build --prod

### STAGE 2:RUN ###
# Defining nginx image to be used
FROM nginx:latest AS ngi
# Copying compiled code and nginx config to different folder
# NOTE: This path may change according to your project's output folder
COPY --from=build /dist/src/app/dist/my-docker-angular-app /usr/share/nginx/html
COPY /nginx.conf /etc/nginx/conf.d/default.conf
# Exposing a port, here it means that inside the container
# the app will be using Port 80 while running
EXPOSE 80
```

Docker Compose





The Concept

- Tool for defining and running multi-container.
- YAML file to configure your application's services.
- Commands for managing the whole lifecycle of your application

Networking

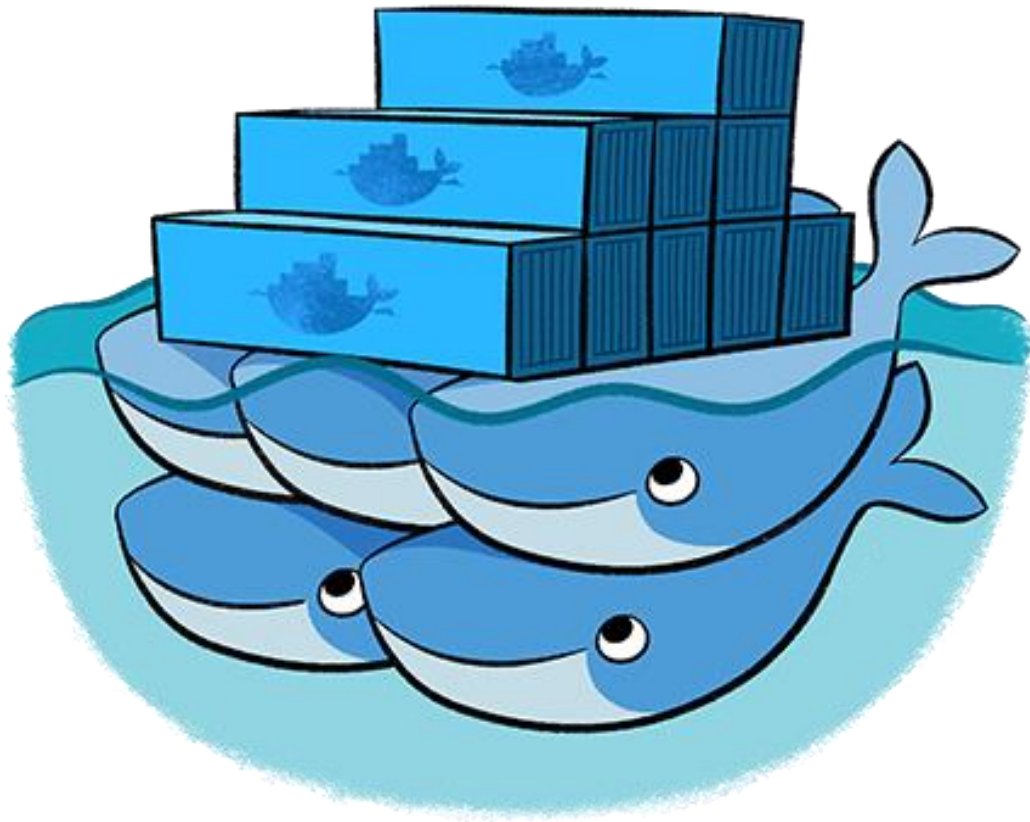
Docker networks are used to help microservices communicate with each other

docker

```
docker run --network my-network -p 8080:80  
-n my_web_container my_web_image:latest
```

docker-compose

```
version: "3.9"  
services:  
  my_web_container:  
    image: my_web_image:latest  
    ports:  
      - "8080:80"  
    networks:  
      - my-network  
networks:  
  my-network:  
    name: my-pre-existing-network  
    external: true  
    driver: true
```

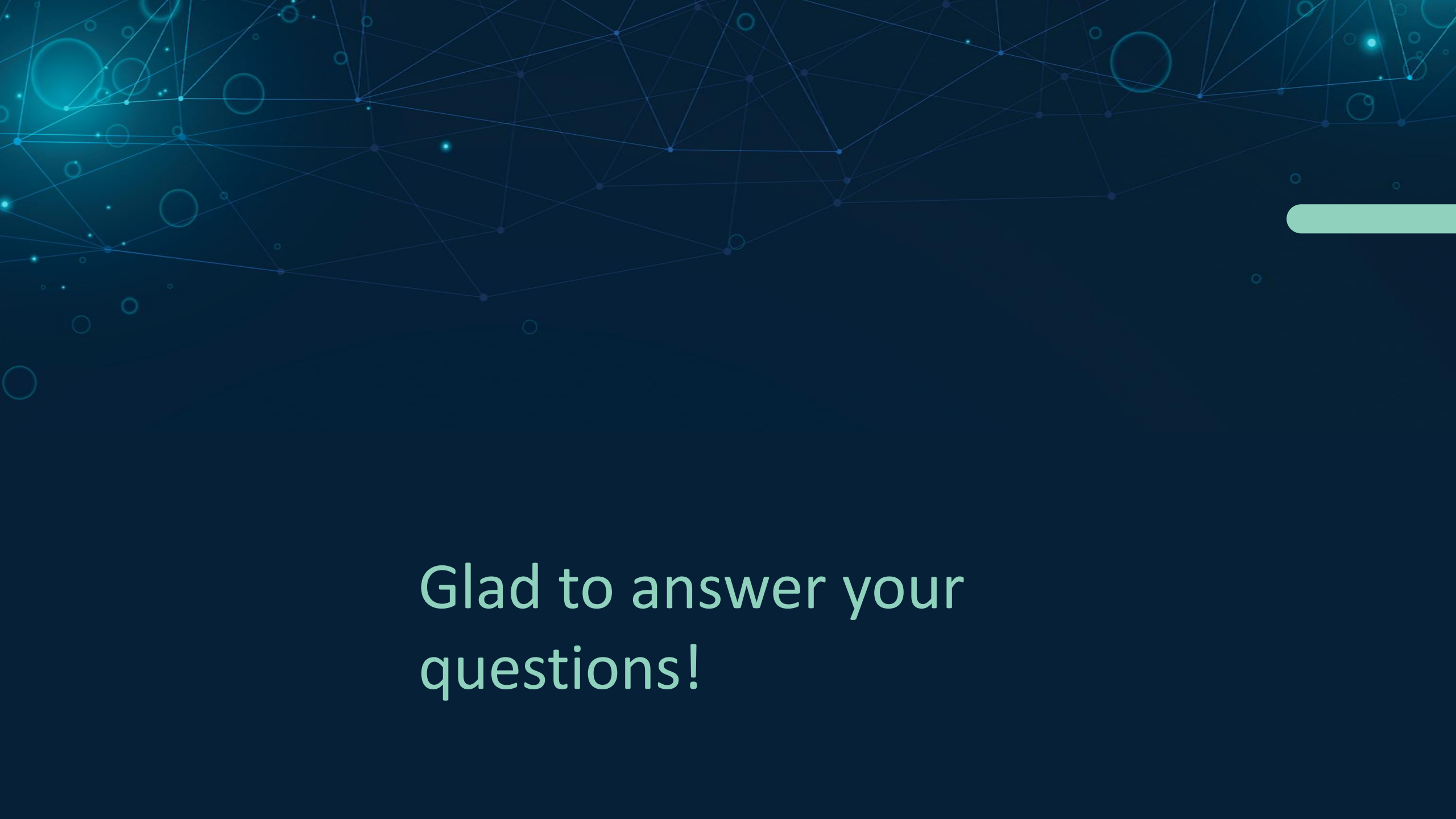


What is Swarm?

- Group of physical or virtual machines running Docker .
- Configured to join together in a cluster.
- Controlled by a swarm manager.
- Machines that have joined the cluster are referred to as nodes
- Allows the user to manage multiple containers deployed across multiple host machines.

Conclusion





Glad to answer your
questions!